

Work through the problems below. Draw pictures where they help — a quick Venn diagram often makes an “and/or/not” question obvious.

Part 1: Set Theory — Unions, Intersections, Complements

1. Roll one fair six-sided die, so $\Omega = \{1, 2, 3, 4, 5, 6\}$. Let A be the event the roll is **even** and B be the event the roll is **4 or greater**.
 - (a) List the outcomes in A and in B .
 - (b) List the outcomes in $A \cup B$.
 - (c) List the outcomes in $A^C \cap B$.
 - (d) List the outcomes in $(A \cup B)^C$.

2. Draw one card from a standard 52-card deck. Let H be the event the card is a **heart** and F be the event the card is a **face card** (Jack, Queen, or King). Write each event below in set notation using H , F , $\cap$, $\cup$, and C .
 - (a) The card is a heart **and** a face card.
 - (b) The card is a heart **but not** a face card.
 - (c) The card is **neither** a heart **nor** a face card.
 - (d) Shade in a Venn diagram to describe the region in part (c).

3. Draw a Venn diagram below for two events A and B inside an outcome space Ω . On separate copies of the diagram:

- (a) Shade the region $A \cap B^C$ (in A but not in B).
- (b) Shade the region $(A \cup B)^C$.
- (c) Shade the outcomes that are in **exactly one** of A or B , and then write that region in set notation.

4. Consider an outcome space Ω with events A, B that are *not* mutually exclusive. Let $P(A) = 0.5$, $P(B) = 0.7$, and $P(A \cap B) = 0.4$.

- (a) Draw a Venn diagram with two circles, one for A and one for B .
- (b) Using your diagram, calculate $P(A \cup B)$. Explain why you cannot simply add $P(A)$ and $P(B)$ to get this answer.
- (c) Find $P(A^C)$.
- (d) Find $P(A \cap B^C)$, the probability of “ A but not B .”

5. For each pair of events below, state whether the two events are **mutually exclusive** (recall this means $A \cap B = \emptyset$).
- (a) Draw one card: “the card is a heart” and “the card is a spade.”
 - (b) Draw one card: “the card is a heart” and “the card is a king.”
 - (c) Roll one die: “the roll is even” and “the roll is odd.”
 - (d) Roll one die: “the roll is even” and “the roll is 4 or greater.”

Part 2: The Axioms of Probability

6. **Inclusion–exclusion.** In a survey, the probability a person drinks coffee is 0.7, the probability they drink tea is 0.4, and the probability they drink **both** is 0.2.
- (a) Define events A = “person drinks coffee” and B = “person drinks tea.” State whether the problem gives you $P(A)$, $P(B)$, $P(A \cap B)$, and $P(A \cup B)$, and write down each value it does give.
 - (b) Find the probability a person drinks coffee **or** tea.
 - (c) Find the probability a person drinks **neither**.
 - (d) Find the probability a person drinks coffee **but not** tea.

7. **Which axiom breaks?** For each proposed assignment of probabilities, decide whether it is *possible*. If it is impossible, name the axiom or consequence it violates and explain.

- (a) $P(A) = 0.6$, $P(B) = 0.7$, and A and B are mutually exclusive.
- (b) $P(A) = -0.2$.
- (c) $P(A) = 0.6$, $P(B) = 0.7$, and $P(A \cap B) = 0.4$.

(BREAK, END OF FIRST PART OF CLASS)

Part 3: Conditional Probability

8. The table below gives the **joint probabilities** for a randomly chosen commuter: their main mode of transport (Bike, Bus, or Car) and whether they arrived **Late** or **On time**. Each entry is the probability of that combination; the margins give the totals.

	Bike	Bus	Car	Total
Late	0.05	0.15	0.05	0.25
On time	0.20	0.25	0.30	0.75
Total	0.25	0.40	0.35	1.00

- (a) Find $P(\text{Late})$ and $P(\text{Bus})$.
- (b) Find $P(\text{Late} \cap \text{Bus})$.
- (c) Find $P(\text{Late} \mid \text{Bus})$.
- (d) Find $P(\text{Bike} \mid \text{Late})$ and $P(\text{Bike} \mid \text{On time})$. Is a commuter more likely to bike when they are late, or when they are on time?
- (e) Is the event “Late” independent of the event “Bus”? Compare $P(\text{Late} \mid \text{Bus})$ to $P(\text{Late})$ to decide.

9. Two events A and B satisfy $P(A) = 0.3$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$.
- (a) Find $P(A \mid B)$. Did conditioning on B make A **more** or **less** likely than its original probability $P(A)$?
 - (b) Find $P(A \mid B^C)$. (Hint: first find $P(A \cap B^C)$ and $P(B^C)$.)
 - (c) Are A and B independent? Explain in a sentence what these numbers say about how A and B are related.
10. Tomorrow there is a 40% chance of rain. If it rains, there is an 80% chance you carry an umbrella; if it does not rain, there is only a 10% chance you carry an umbrella.
- (a) Find $P(\text{rain and umbrella})$ using the rule $P(A \cap B) = P(B) P(A \mid B)$.
 - (b) Find $P(\text{no rain and umbrella})$.
 - (c) Use parts (a) and (b) to find $P(\text{umbrella})$. (Hint: “rain” and “no rain” are mutually exclusive, so add the two pieces.)